Homework 1: Volatility & Correlation

Asset Allocation — FRE 6991, Fall 2026

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Prepared November 3, 2026 • Due Sunday, November 8, 2026

1 Data and measurement conventions

The assignment considers Apple (AAPL), Exxon Mobil (XOM), Pfizer (PFE), and Consolidated Edison (ED). This report calculates their daily returns for October 1, 2024 through October 1, 2026, inclusive; estimates 30-trading-day rolling volatility and all six pairwise correlations; and constructs a mean–variance efficient frontier from monthly returns.

Daily historical observations were obtained from Yahoo Finance through `yfinance` and frozen as four raw CSV files on November 3, 2026. The analysis uses the **adjusted close**, which accounts for splits and cash dividends, rather than the unadjusted closing price. Yahoo's historical pages and adjusted-close explanation are listed in the references. The provided assignment is retained in the repository.

Table 1: Sample boundaries and observation counts.

Item	Convention / observations
Price observations per stock	503, September 30, 2024–October 1, 2026
Daily return observations per stock	502, October 1, 2024–October 1, 2026
Rolling window	30 trading-day returns; 473 complete windows
Rolling window ending dates	November 11, 2024–October 1, 2026
Monthly return observations per stock	24, October 2024–September 2026
Annualization factors	252 trading days; 12 months
Sample standard deviations / covariance	Denominator $n-1$ (`ddof=1`)

All stocks have identical trading dates, positive adjusted prices, and no missing observations. No dates are dropped and no prices are forward-filled. September 30, 2024 supplies the prior close needed to calculate the first requested daily return. October 1, 2026 is included in the daily analysis; the incomplete October 2026 month is excluded from monthly estimation.

For stock i, let $P^*_{i,t}$ denote its adjusted close. The simple daily return is

$$r_{i,t} = \frac{P^*_{i,t}}{P^*_{i,t-1}} - 1. \tag{1}$$

These are simple returns rather than logarithmic returns. Calculations use unrounded decimal returns throughout. Tables explicitly report percentages, dimensionless correlations, or covariance in squared decimal-return units. The complete daily return series is supplied in `results/daily_returns.csv`; selected rows appear on the next page.

2 Daily returns

Table 2: First and last three daily returns (percent). The CSV contains all 502 dates.

Date	AAPL	XOM	PFE	ED
2024-10-01	-2.9142%	2.3119%	-0.9330%	0.9795%
2024-10-02	0.2520%	1.3258%	-0.2093%	-0.6657%
2024-10-03	-0.4895%	0.8723%	-0.9437%	-0.4117%
2026-09-29	-2.6596%	-0.7199%	0.0000%	0.4862%
2026-09-30	1.0990%	0.8677%	-0.6964%	-1.0839%
2026-10-01	-0.8108%	0.6575%	-1.4025%	1.2621%

Table 3: Full daily-sample statistics (percent). Annualized volatility is $\sqrt{252}$ times daily sample standard deviation.

	Mean daily	Daily std.	Annualized vol.	Cumulative return
AAPL	0.09%	1.82%	28.83%	42.95%
XOM	0.09%	1.56%	24.81%	49.00%
PFE	0.03%	1.51%	23.95%	11.28%
ED	0.02%	1.17%	18.51%	6.27%

The cumulative return is $\prod_t (1 + r_{i,t}) - 1$, covering the September 30, 2024 close through the October 1, 2026 close. XOM has the largest cumulative return, 49.00%, followed by AAPL at 42.95%. AAPL has the greatest full-sample daily-based annualized volatility, 28.83%; ED has the smallest, 18.51%. The cumulative returns and mean daily returns measure different quantities: the former compound the entire sample, whereas the latter are arithmetic averages.

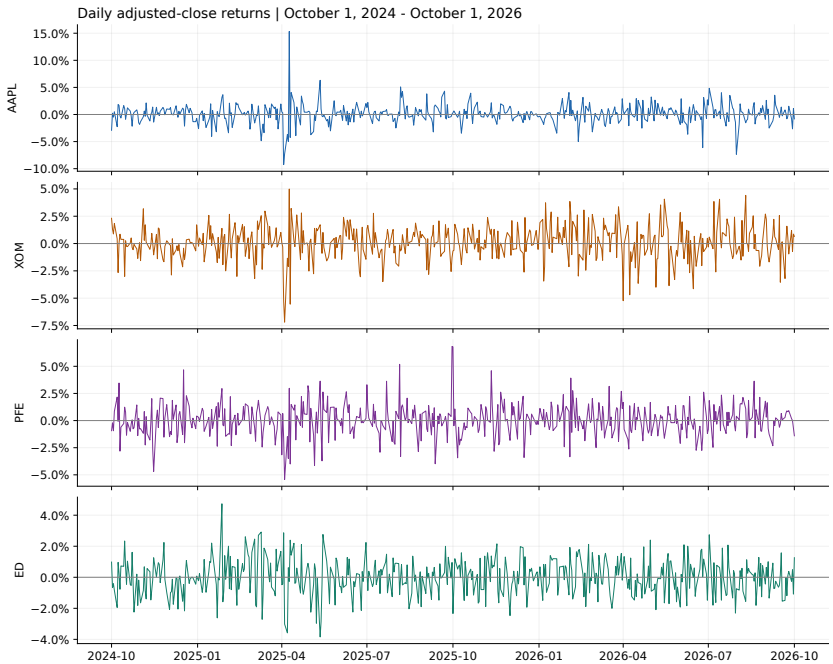


Figure 1: The four daily adjusted-close return series. The panels share dates and express returns in percent; vertical ranges vary by stock.

3 Thirty-day rolling annualized volatility

For each complete window ending at trading date t, define

$$\bar{r}_{i,t}^{(30)} = \frac{1}{30}\sum_{k=0}^{29} r_{i,t-k}, \qquad \widehat{\sigma}_{i,t}^{\text{ann}} = \sqrt{252}\sqrt{\frac{1}{29}\sum_{k=0}^{29}\left(r_{i,t-k} - \bar{r}_{i,t}^{(30)}\right)^2}. \tag{2}$$

The window contains **30 trading days**, and advances by one trading day. Thus there are $502-30+1 = 473$ volatility estimates for each stock. The square-root-of-time conversion assumes a common daily variance scale without serial covariance; it is a convention for comparison, not a forecast.

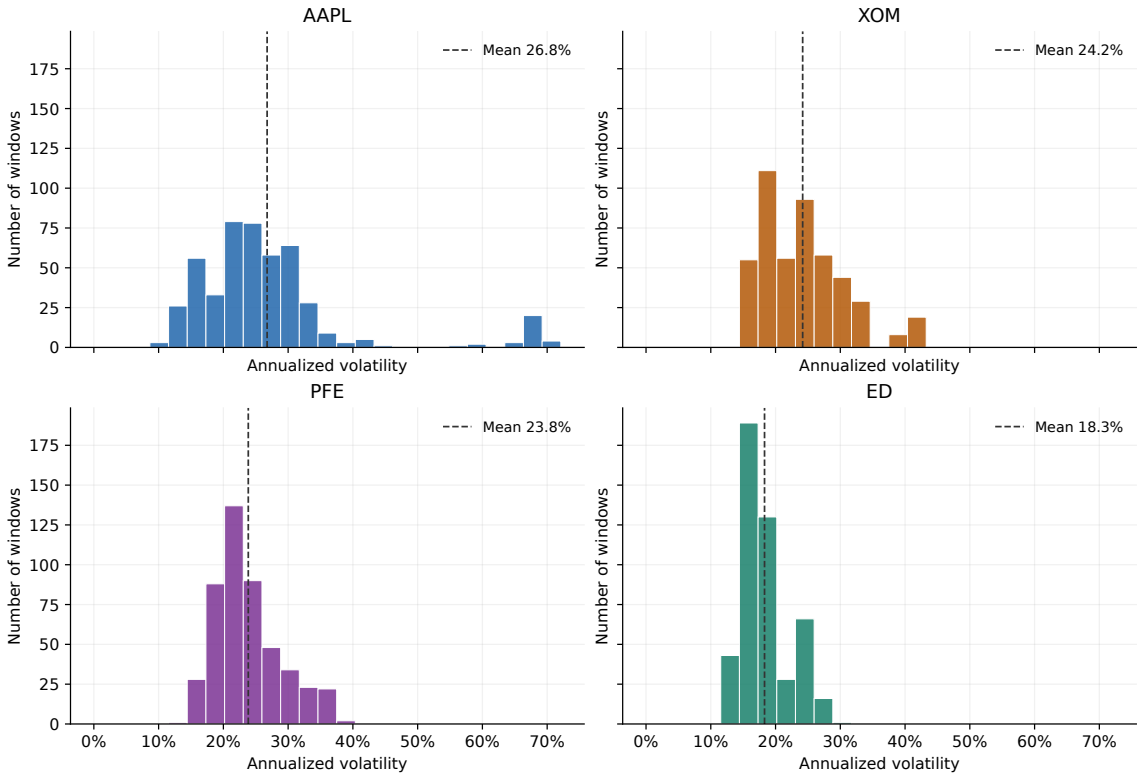


Figure 2: Four histograms of 30-day rolling annualized volatility, one for each stock. Common bin boundaries and axes are used. Dashed lines mark the means of the 473 estimates.

Table 4: Distribution of rolling annualized volatility (percent; the standard-deviation column measures variation across windows).

	mean	std	min	median	max
AAPL	26.76%	12.13%	10.24%	25.11%	70.01%
XOM	24.19%	6.31%	14.48%	23.75%	41.77%
PFE	23.85%	5.09%	14.36%	22.79%	38.03%
ED	18.30%	3.63%	12.30%	17.36%	29.15%

AAPL's distribution has the widest spread: rolling annualized volatility ranges from 10.24% to 70.01%, with a 26.76% mean. ED has the lowest mean, 18.30%, and the smallest standard deviation across windows, 3.63 percentage points. XOM and PFE have similar mean rolling volatility, 24.19% and 23.85%. The mean rolling volatility need not equal the full-sample volatility in Table 3, because averaging local standard deviations differs from taking one standard deviation over all dates.

4 Thirty-day rolling correlations

For each unordered pair i, j and the same window, calculate Pearson correlation:

$$\widehat{\rho}_{ij,t}^{(30)} = \frac{\sum_{k=0}^{29}(r_{i,t-k} - \bar{r}_{i,t}^{(30)})(r_{j,t-k} - \bar{r}_{j,t}^{(30)})}{\sqrt{\sum_{k=0}^{29}(r_{i,t-k} - \bar{r}_{i,t}^{(30)})^2}\sqrt{\sum_{k=0}^{29}(r_{j,t-k} - \bar{r}_{j,t}^{(30)})^2}}. \tag{3}$$

Correlation is dimensionless and is not annualized. There are $\binom{4}{2} = 6$ pairs and 473 estimates per pair.

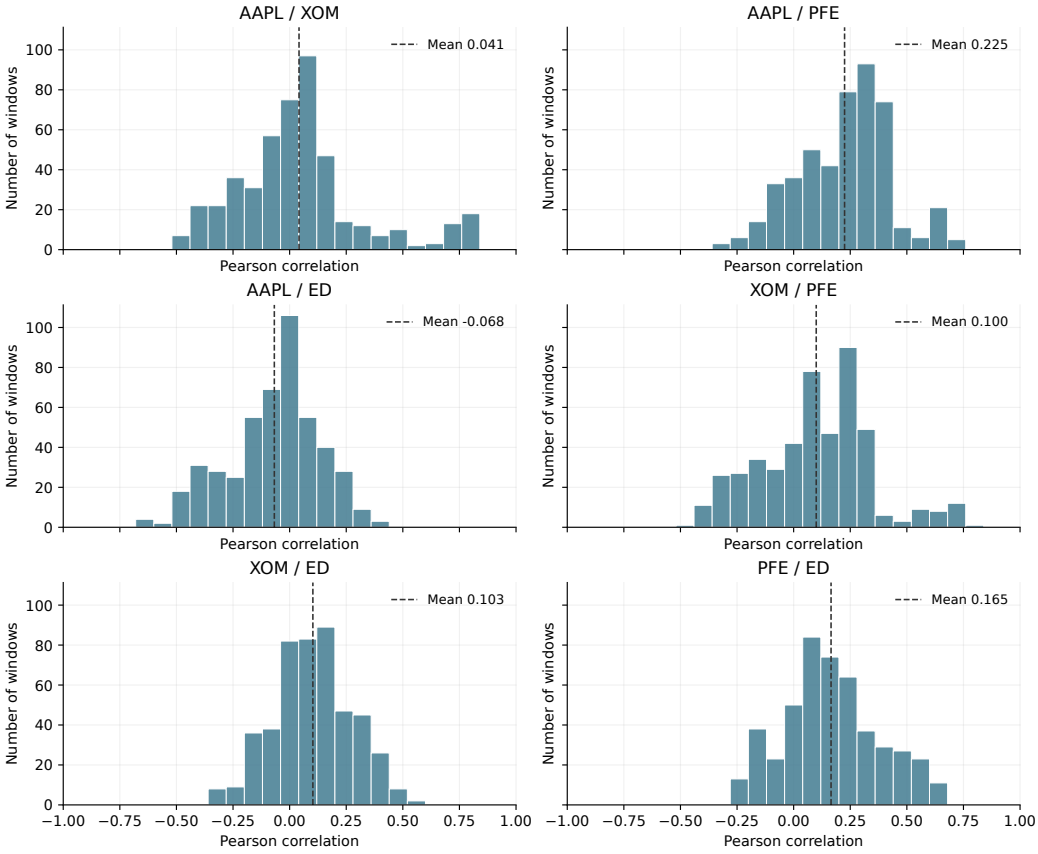


Figure 3: All six histograms of 30-day rolling correlations, using common bins from −1 to 1. Dashed lines mark the mean correlation for each pair.

Table 5: Rolling correlation summaries (dimensionless).

	mean	std	min	median	max
AAPL / XOM	0.041	0.279	-0.463	0.032	0.808
AAPL / PFE	0.225	0.203	-0.328	0.259	0.694
AAPL / ED	-0.068	0.205	-0.636	-0.035	0.408
XOM / PFE	0.100	0.249	-0.441	0.106	0.761
XOM / ED	0.103	0.174	-0.348	0.102	0.538
PFE / ED	0.165	0.209	-0.273	0.145	0.665

AAPL/PFE has the highest mean rolling correlation, 0.225. AAPL/ED is the only pair with a negative mean, −0.068. All six pairs take both negative and positive values during the sample, so a single estimate conceals substantial variation. Adjacent windows share 29 observations: these histogram counts describe overlapping estimates and must not be treated as independent draws.

5 Monthly return statistics

Let $P^*_{i,m}$ be the adjusted close on the last observed trading day of month m. Monthly returns compound the daily returns in that month:

$$R_{i,m} = \frac{P^*_{i,m}}{P^*_{i,m-1}} - 1 = \prod_{t \in m}(1 + r_{i,t}) - 1, \qquad m = \text{October 2024}, \ldots, \text{September 2026}. \tag{4}$$

The September 30, 2024 baseline and the last trading days of the following 24 complete months give 24 monthly returns. A month-end label in the CSV can fall on a non-trading day; the price is the last available trading-day close. The monthly ratios were checked against compounded daily returns.

With $M = 24$, the monthly arithmetic mean and sample covariance are

$$\widehat{\mu}_{i,m} = \frac{1}{M}\sum_{m=1}^{M} R_{i,m}, \qquad (\widehat{\Sigma}_m)_{ij} = \frac{1}{M-1}\sum_{m=1}^{M}(R_{i,m} - \widehat{\mu}_{i,m})(R_{j,m} - \widehat{\mu}_{j,m}). \tag{5}$$

The frontier uses $\widehat{\mu}_a = 12\widehat{\mu}_m$ and $\widehat{\Sigma}_a = 12\widehat{\Sigma}_m$, hence $\widehat{\sigma}_{i,a} = \sqrt{12}\widehat{\sigma}_{i,m}$. Annual means are annualized *arithmetic* means, not compounded growth rates. Covariance scaling assumes serially uncorrelated monthly returns.

Table 6: Monthly statistics and annualized inputs from 24 monthly returns (percent).

	Monthly mean	Monthly vol.	Annual mean	Annual vol.
AAPL	1.72%	6.24%	20.60%	21.61%
XOM	1.88%	7.02%	22.55%	24.31%
PFE	0.63%	5.07%	7.51%	17.56%
ED	0.33%	5.14%	3.93%	17.79%

Table 7: Monthly return correlation matrix (dimensionless).

	AAPL	XOM	PFE	ED
AAPL	1.0000	-0.1402	0.2539	-0.4491
XOM	-0.1402	1.0000	0.5059	0.3911
PFE	0.2539	0.5059	1.0000	-0.0061
ED	-0.4491	0.3911	-0.0061	1.0000

Table 8: Monthly sample covariance matrix (squared decimal-return units, not percent). Annualized covariance is this matrix multiplied by 12.

	AAPL	XOM	PFE	ED
AAPL	0.003892	-0.000614	0.000803	-0.001439
XOM	-0.000614	0.004925	0.001800	0.001410
PFE	0.000803	0.001800	0.002571	-0.000016
ED	-0.001439	0.001410	-0.000016	0.002638

XOM has the largest annualized mean, 22.55%, and the largest monthly-based annualized volatility, 24.31%. PFE has the smallest annualized volatility, 17.56%. The strongest positive monthly correlation is XOM/PFE, 0.5059; AAPL/ED is the most negative, -0.4491, and contributes diversification. Monthly correlations and average rolling daily correlations differ because aggregation and the estimation windows differ. These are separate estimates.

6 Efficient frontier: formulation and portfolios

The primary analysis assumes a fully invested portfolio with no short sales:

$$\mathbf{1}^T w = 1, \qquad w_i \geq 0, \qquad \mu_p = w^T \widehat{\mu}_a, \qquad \sigma_p = \sqrt{w^T \widehat{\Sigma}_a w}. \tag{6}$$

No transaction costs, taxes, or risk-free asset are included. A risk-free rate is not specified in the assignment, so a tangency portfolio or capital allocation line is not calculated.

First, solve the global minimum-variance problem with the budget and nonnegativity constraints. Then, for each target annual arithmetic return q from the minimum-variance portfolio's mean through the largest asset mean, solve

$$\min_w w^T \widehat{\Sigma}_a w \quad \text{subject to} \quad \mathbf{1}^T w = 1, \quad \widehat{\mu}_a^T w = q, \quad w \geq 0. \tag{7}$$

This upper branch is the efficient frontier: it gives the smallest volatility available at each efficient expected return. The lower-return branch below the minimum-variance portfolio's mean is not efficient.

Exact constrained solution

With four assets, all $2^4 - 1 = 15$ possible nonempty supports can be examined. For a support S, let C_S be its annual covariance submatrix, and define

$$H_S = \begin{pmatrix} \mathbf{1}_S^T \\ \widehat{\mu}_{a,S}^T \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ q \end{pmatrix}, \qquad w_S = C_S^{-1} H_S^T \left(H_S C_S^{-1} H_S^T \right)^{-1} b. \tag{8}$$

Feasible nonnegative solutions are retained and their variances compared; single-asset supports are handled directly. For the global minimum-variance problem, $H_S = \mathbf{1}_S^T$ and $b = 1$. The covariance matrix is positive definite, so this procedure finds the global constrained solution. The graph samples 201 targets; each target is solved analytically over the possible supports. Random portfolios illustrate the feasible region, and do not determine the frontier.

Table 9: Selected fully invested long-only portfolios. Weights, annualized arithmetic means, and annualized volatilities are percentages.

	AAPL	XOM	PFE	ED	Annual mean	Annual vol.
Minimum variance	33.05%	0.00%	19.76%	47.19%	10.15%	9.59%
Equal weight	25.00%	25.00%	25.00%	25.00%	13.65%	11.59%
Midpoint target	46.58%	24.99%	0.00%	28.43%	16.35%	11.18%
Maximum mean	0.00%	100.00%	0.00%	0.00%	22.55%	24.31%

The minimum-variance allocation is **33.05% AAPL, 0.00% XOM, 19.76% PFE, and 47.19% ED**. Its estimated annual mean is 10.15%, and its annual volatility is 9.59%, below the volatility of every individual stock. The midpoint target is halfway between the minimum-variance mean and the largest asset mean, not a portfolio with equal weights. It has 16.35% mean and 11.18% volatility, compared with 13.65% and 11.59% for equal weights. Thus equal weights are inefficient under these sample estimates. The highest feasible long-only mean, 22.55%, is obtained by holding 100% XOM.

7 Efficient frontier: graph and interpretation

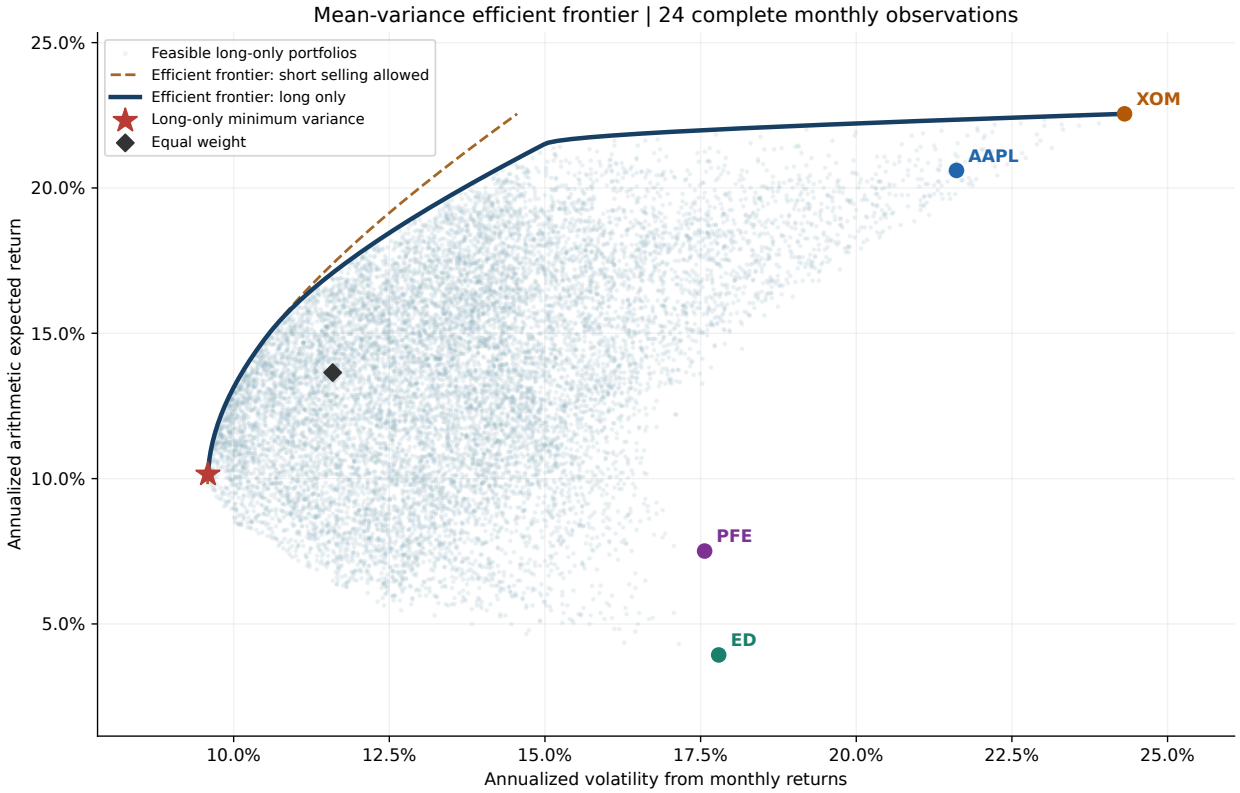


Figure 4: Efficient frontiers using annualized moments estimated from the 24 monthly returns. The solid line is the long-only upper frontier; the dashed line permits short sales. The cloud contains 12,000 illustrative long-only portfolios. Individual stocks, the long-only minimum-variance portfolio, and equal weights are marked.

The leftmost point of the long-only frontier is the minimum-variance portfolio. Moving upward along the curve requires increasing risk to achieve larger estimated returns. A substantial allocation to ED can reduce portfolio variance despite ED's low estimated mean, in part because its monthly returns correlate negatively with AAPL. The efficient allocation changes with the target return; minimum standalone volatility alone does not determine portfolio weights.

Supplemental frontier with short selling permitted

Removing $w \geq 0$ retains the fully invested budget and target-return constraints. Define, using the full annual covariance matrix $C = \widehat{\Sigma}_a$,

$$a = \mathbf{1}^T C^{-1}\mathbf{1}, \quad b = \mathbf{1}^T C^{-1}\widehat{\mu}_a, \quad c = \widehat{\mu}_a^T C^{-1}\widehat{\mu}_a, \quad d = ac - b^2. \tag{9}$$

The unrestricted minimum variance at target q is

$$\sigma^2(q) = \frac{aq^2 - 2bq + c}{d}, \qquad w(q) = C^{-1}\left(\frac{c - bq}{d}\mathbf{1} + \frac{aq - b}{d}\widehat{\mu}_a\right). \tag{10}$$

Its efficient branch starts at $q = b/a$. The unrestricted minimum-variance portfolio has approximately 32.75% AAPL, −1.83% XOM, 21.10% PFE, and 47.98% ED, with 9.81% annual mean and 9.58% annual volatility. Removing nonnegativity weakly lowers minimum risk for a given feasible target. For comparison, the dashed curve is displayed only through XOM's estimated mean. Unrestricted feasible target returns extend beyond that displayed range; XOM's mean is an upper bound only for long-only portfolios.

8 Reproducibility and interpretation limits

The repository includes raw data, calculation code, full-precision CSV results, an Excel workbook, chart files, the LaTeX source, and this report. All paths below are relative to the repository root.

Table 10: Principal calculation artifacts. CSV returns and weights use decimals.

Artifact	Path
Original assignment	`assignment/HW1_VOLATILITYANDCORRELATION1.docx`
Frozen adjusted-price source files	`data/raw/AAPL.csv`, `XOM.csv`, `PFE.csv`, `ED.csv` in the same directory
Download metadata	`data/raw/download_metadata.json`
Reproducible analysis	`scripts/analyze.py`
All daily returns	`results/daily_returns.csv`
All rolling volatility estimates	`results/rolling_30d_annualized_volatility.csv`
All six rolling correlations	`results/rolling_30d_correlations.csv`
Monthly returns and moments	`results/monthly_returns.csv`, `monthly_summary.csv`, `monthly_correlation.csv`, `monthly_covariance.csv` in `results/`
Annual covariance for optimization	`results/annualized_monthly_covariance.csv`
Frontier returns, risk, and weights	`results/efficient_frontier_long_only.csv`; `results/efficient_frontier_unconstrained.csv`
Excel calculation workbook	`results/homework1_calculations.xlsx`
Validation and source-file hashes	`results/summary.json`

Run `python scripts/analyze.py` from the repository root after installing the dependencies in `requirements.txt`. The frozen CSV inputs allow reproduction without downloading prices again. Generated LaTeX tables and PDF charts are read directly into this report, reducing transcription error. Checks verify aligned dates, no missing prices, the number of rolling windows, monthly compounding, symmetric correlations, positive-definite covariance, portfolio budgets, target returns, and nonnegative long-only weights. An independent constrained numerical optimizer agrees with selected analytical frontier solutions to well within 10^{-8} in annual variance.

The frontier is an *in-sample* estimate from just 24 monthly observations. Estimated means, covariance, and portfolio weights can change materially with the sample period. The optimization does not establish future returns or a uniquely preferred portfolio; that choice also requires an investor's risk preferences. Daily-based and monthly-based volatility estimates need not coincide, since their frequencies and endpoints differ. Rolling windows overlap, so their distributions provide descriptive evidence about changing risk and dependence rather than 473 independent observations.

References

[1] Asset Allocation (FRE 6991), Fall 2026. *Homework 1: Volatility & Correlation.* Supplied assignment, `assignment/HW1_VOLATILITYANDCORRELATION1.docx`.

[2] Yahoo Finance. *Apple Inc. (AAPL): Historical Data.* `https://finance.yahoo.com/quote/AAPL/history/`.

[3] Yahoo Finance. *Exxon Mobil Corporation (XOM): Historical Data.* `https://finance.yahoo.com/quote/XOM/history/`.

[4] Yahoo Finance. *Pfizer Inc. (PFE): Historical Data.* `https://finance.yahoo.com/quote/PFE/history/`.

[5] Yahoo Finance. *Consolidated Edison, Inc. (ED): Historical Data.* `https://finance.yahoo.com/quote/ED/history/`.

[6] Yahoo Help. *What is the adjusted close?* `https://in.help.yahoo.com/kb/adjusted-close-sln28256.html`. Historical observations retrieved through `yfinance` on November 3, 2026.